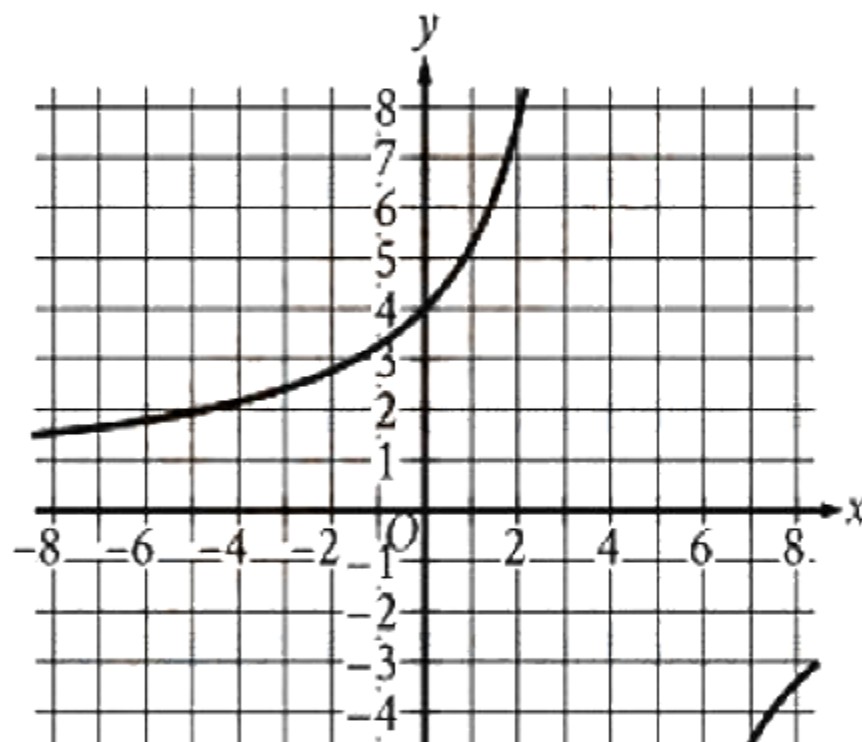


1



Mark for Review



The graph of $y = f(x)$ is shown in the xy -plane. What is the value of $f(0)$?

- A) -4
- B) 0
- C) $\frac{3}{100}$
- D) 4

2



Mark for Review



Which expression is equivalent to $(7nr^2 + 2nr) + (5n^2r + 4nr)$?

(A) $12n^2r^2 + 6nr$

(B) $12n^3r^3 + 6n^2r^2$

(C) $35n^3r^3 + 8n^2r^2$

(D) $7nr^2 + 5n^2r + 6nr$



$$v = -\frac{w}{123x}$$

The given equation relates the distinct positive numbers v , w , and x . Which equation correctly expresses w in terms of v and x ?

(A) $w = -123vx$

(B) $w = -\frac{123v}{x}$

(C) $w = -\frac{x}{123v}$

(D) $w = v + 123x$



$$-3x + 6rx = 54$$

In the given equation, r is a constant. The equation has no solution. What is the value of r ?

(A) $\frac{1}{9}$

(B) $\frac{1}{2}$

(C) 2

(D) 9

Student-produced response directions

- If you find **more than one correct answer**, enter only one answer.
- You can enter up to 5 characters for a **positive** answer and up to 6 characters (including the negative sign) for a **negative** answer.
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- Don't enter **symbols** such as a percent sign, comma, or dollar sign.

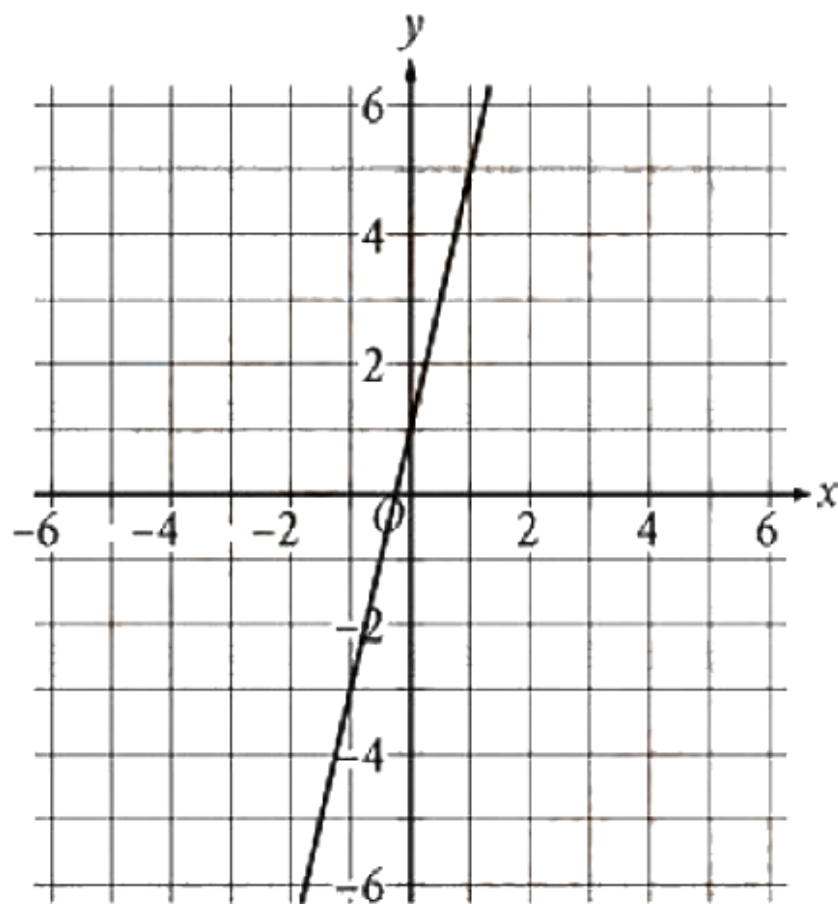


5



Mark for Review

A real estate company offers a series of three webinars. 3,750 people attended the first webinar. 54% of the people who attended the first webinar attended the second webinar, and 36% of the people who attended the first and second webinars attended the third webinar. How many people attended all three webinars?



The graph of $y = f(x)$ is shown. What is the y -intercept of the graph of $y = f(x) + 11$?

- A. (0,-10)
- B. (0,12)
- C. (1,11)
- D. (-11,1)



The function f is defined by $f(x) = |x - 6x|$. What value of a satisfies $f(2) - f(a) = -30$?

(A) -32

(B) 6

(C) 8

(D) 150

Student-produced response directions

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8



Mark for Review

If $-7(13 - 5x) + 3 = -8(13 - 5x) + 25$, what is the value of $13 - 5x$?

Answer Preview:



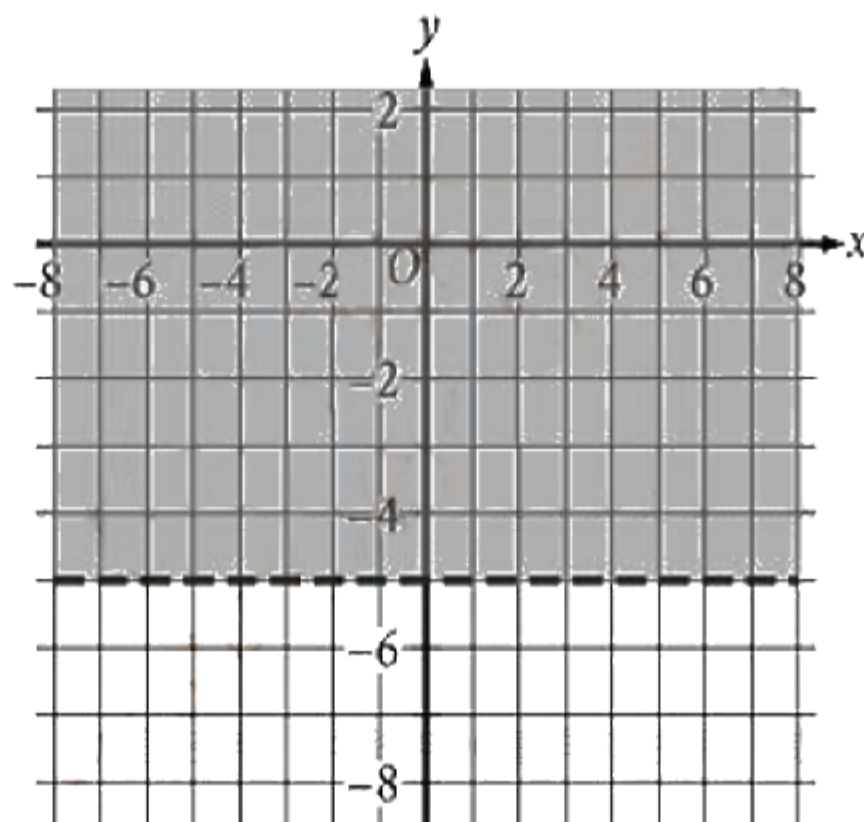
The minimum value of x is 4 less than 8 times another number n . Which inequality shows the possible values of x ?

(A) $x \leq 8n - 4$

(B) $x \geq 8n - 4$

(C) $x \leq 4 - 8n$

(D) $x \geq 4 - 8n$



The shaded region shown represents the solutions to the inequality $-19y < c$, where c is a constant. What is the value of c ?

- A. 95
- B. 5
- C. -5
- D. -95



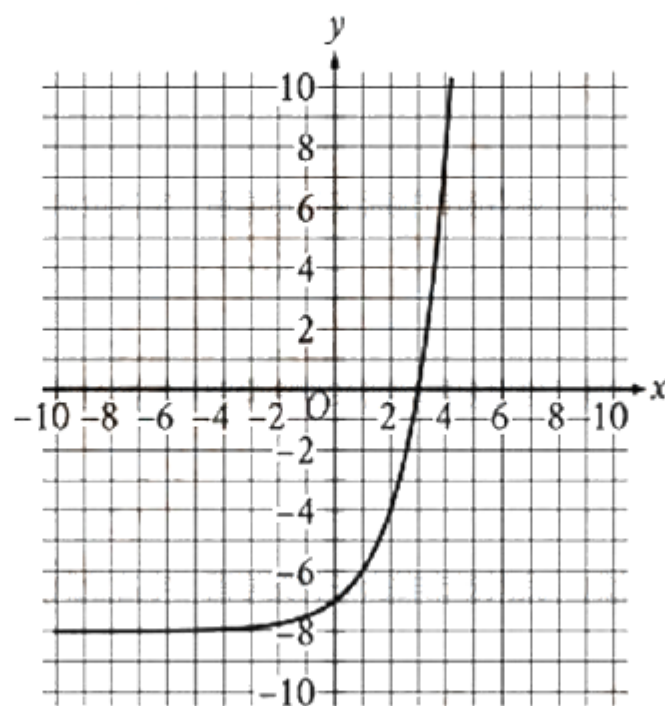
At a constant temperature, a scuba diver combines two different concentrations of nitrox mix (oxygen-nitrogen gas mixture) to fill a tank to a total pressure of 90 atmospheres (atm), of which 37% is contributed by oxygen. Nitrox mix A contributes x atm of pressure to the tank, of which 22% is contributed by oxygen. Nitrox mix B contributes y atm of pressure to the tank, of which 49% is contributed by oxygen. Which system of equations represents this situation?

(A) $x + y = 90$
 $0.22x + 0.49y = 0.37(90)$

(B) $x + y = 90$
 $0.49x + 0.22y = 0.37(90)$

(C) $x + y = 0.37(90)$
 $0.22x + 0.49y = 90$

(D) $x + y = 0.37(90)$
 $0.49x + 0.22y = 90$



- The graph of the equation $y = 2^x + k$ is shown, where k is a constant. What is the value of k ?
- A. -8
 - B. -7
 - C. 7
 - D. 8

13



Mark for Review



A certain town has an area of 3.51 square miles. What is the area, in square yards, of this town? (1 mile = 1,760 yards)

(A) 501

(B) 6,178

(C) 882,507

(D) 10,872,576



The function f gives the product of a number, x , and a number that is 7 more than x . Which equation defines f ?

(A) $f(x) = x^2 + x + 7$

(B) $f(x) = x^2 + 7$

(C) $f(x) = x^2 + 7x$

(D) $f(x) = x^2 + 7x + 7$



The function $f(x) = \frac{1}{8}(x - 6)^2 + 4$ gives a toy car's height above the ground $f(x)$, in inches, x seconds after it started moving on an elevated track, where $0 \leq x \leq 10$. Which of the following is the best interpretation of the vertex of the graph of $y = f(x)$ in the xy -plane?

- (A) The toy car's minimum height was 4 inches above the ground.
- (B) The toy car's minimum height was 6 inches above the ground.
- (C) The toy car's height was 4 inches above the ground when it started moving.
- (D) The toy car's height was 6 inches above the ground when it started moving.

Student-produced response directions

- If you find **more than one correct answer**, enter only one answer.
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16



Mark for Review

One of the factors of $5x^3 + 105x^2 + 540x$ is $x + b$, where b is a positive constant. What is the smallest possible value of b ?

Answer Preview:



Kenny engages in up to 3 types of exercise each week for a total of 6 hours while training for a triathlon. Kenny swims for the same number of minutes each week. The equation $y = 360 - x - 170$ represents the situation where Kenny bikes for x minutes during a week and runs for any remaining training time y , in minutes. If this equation is graphed in the xy -plane, which statement is the best interpretation of the x -intercept of the graph?

- (A) During a week when Kenny doesn't run, he bikes for 190 minutes.
- (B) During a week when Kenny doesn't bike, he runs for 190 minutes.
- (C) Each week, Kenny bikes and runs for a total of 190 minutes.
- (D) During a week when Kenny swims for 190 minutes, he bikes for 170 minutes.



Two lines intersect at exactly one point, forming two acute angles and two obtuse angles. The measure of one of these angles is $(9x - 110)^\circ$. Which of the following could NOT be the sum of the measures of any two of these angles?

(A) $(-18x + 220)^\circ$

(B) $(-18x + 580)^\circ$

(C) $(18x - 220)^\circ$

(D) 180°

Student-produced response directions

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**19**

Mark for Review

$$\frac{5}{8}(5x + 8)(x + \sqrt{5k + 8})(x - \sqrt{5k + 8}) = 0$$

In the given equation, k is a positive constant. The product of the solutions to the equation is 64. What is the value of k ?

Answer Preview:

20. Each of the following frequency tables represents a data set. Which of these frequency tables represents the data set with the smallest standard deviation?

A

Value	Frequency
4	0
5	10
6	40
7	10
8	0

C

Value	Frequency
4	12
5	12
6	12
7	12
8	12

B

Value	Frequency
4	8
5	14
6	16
7	14
8	8

D

Value	Frequency
4	15
5	11
6	8
7	11
8	15

Student-produced response directions

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Examples

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21

Mark for Review

	Volume (cubic units)
Right circular cylinder A	72π
Right circular cylinder B	$4,608\pi$

The table shows the volume of two similar solids, right circular cylinder A and right circular cylinder B. The radius of right circular cylinder A is 3 units. The surface area of right circular cylinder A is $k\pi$ square units, and the surface area of right circular cylinder B is $n\pi$ square units, where k and n are constants. What is the value of $n - k$? (The surface area of a right circular cylinder with radius r and height h is $2\pi r^2 + 2\pi rh$.)

Student-produced response directions

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22

Mark for Review

Function f is a quadratic function. The graph of $y = f(x)$ in the xy -plane has a vertex at $(-5, -8)$, contains the point $(-6, -10)$, and has a y -intercept at $(0, a)$. The graph of $y = 6 \cdot f(x)$ has a y -intercept at $(0, b)$. What is the positive difference between a and b ?

Answer Preview: